

 Part 1 – Question Bank**10-Mark Questions**

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2. Explain the OSI Security Architecture in detail, focusing on its three main classifications.
3. Provide a comprehensive overview of the Symmetric Cipher Model, its five essential ingredients, and two fundamental requirements.
4. Detail the Playfair Cipher, including its matrix generation and the four specific rules for encryption.
5. Examine the Hill Cipher technique, focusing on its mathematical basis, encryption, and decryption processes.

5-Mark Questions

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7. Identify and describe the four types of computer security as defined in the sources.
8. Explain the "Model for Network Security" and the four basic tasks involved in designing a security service.
9. Discuss the Caesar Cipher, its mathematical algorithm, and its primary advantages.
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13. Describe the Rail Fence technique as a form of transposition cipher.
14. Explain the different types of cyber attacks, specifically focusing on Malware and Social Engineering.
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2-Mark Questions

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25. What is "Spoofing"?

EXAM POV NOTES (Q&A FORMAT)**✦ Part 2 – Complete Answer Bank****10-Mark Questions****Q1. Analyze the three key objectives of computer security (CIA Triad) and their related concepts.**

Introduction: Computer security is defined as the practice of protecting and securing computers, networks, and data from unauthorized access, misuse, or theft. It focuses on three core objectives known as the CIA triad: Confidentiality, Integrity, and Availability.

- **Data Confidentiality Concept:** This assures that private or sensitive information is not made available or disclosed to unauthorized individuals. It is a foundational requirement for protecting proprietary information from being viewed by those without permission. Maintaining this prevents the unauthorized disclosure of information which could lead to various security risks.
- **Privacy Objectives:** Privacy ensures that individuals maintain control or influence over what information related to them is collected and stored. It also dictates who is permitted to disclose that information to third parties. This concept is critical in the modern era where digital footprints are extensive and personal data is highly valuable.
- **Data Integrity Assurance:** Data integrity ensures that information and programs are changed only in a specified and authorized manner. This applies to both data stored on local systems and information transmitted in packets across a network. It guards against improper modification or destruction of information during its lifecycle.
- **System Integrity Purpose:** System integrity assures that a system performs its intended function in an unimpaired manner. This means the system must be free from deliberate or inadvertent unauthorized manipulation. It ensures that the operational logic of the computer remains trustworthy and stable.
- **Availability Goal:** Availability ensures timely and reliable access to and use of information and information systems for authorized users. A loss of availability is often characterized by the disruption of access or the denial of service. Security measures must guarantee that systems work promptly when requested by legitimate entities.
- **Authenticity Requirements:** Authenticity involves ensuring that a message or communication is genuine and comes from the claimed source. It is closely linked to integrity, as it verifies that the data has not been modified by a third party. This objective is often achieved through the use of digital signatures or codes.
- **Non-repudiation Service:** Non-repudiation prevents either the sender or the receiver from denying a transmitted message. It uses techniques to create a verifiable record of the origin and transmission of data. This is essential for resolving disputes regarding whether a specific transaction actually occurred.
- **Unauthorized Access Risks:** Unauthorized access poses a significant threat to all three objectives of the CIA triad. Hackers and thieves often target systems for monetary gains, ransom, or to bully others. Protecting against these risks is the primary reason why computer security is considered essential today.
- **Impact of Loss of Confidentiality:** A loss of confidentiality occurs when there is unauthorized disclosure of information. This can result in the exposure of personal privacy or proprietary business secrets. Such breaches often lead to reputational damage or significant financial losses for organizations.

- **Impact of Loss of Integrity:** A loss of integrity involves the unauthorized modification or destruction of critical information. This can lead to system failures, corrupted databases, or the execution of malicious logic. Ensuring integrity is vital for the reliability of information-driven decision-making.

Q2. Explain the OSI Security Architecture in detail, focusing on its three main classifications.

Introduction: The OSI Security Architecture provides a systematic framework for defining security requirements and approaches to satisfy them. Established by the ISO, it focuses on security attacks, mechanisms, and services.

- **Systematic Framework:** The architecture provides a conceptual framework for understanding how networking protocols interact within a layered structure. It defines security at each of the seven layers of the OSI model. This ensures a flow of safety and security that is internationally acceptable for organizations.
- **Security Attack Definition:** A security attack is defined as any action that compromises the security of information owned by an organization. These are attempts by entities to gain unauthorized access or disrupt system operations. Attacks are categorized into passive and active based on their behavior.
- **Passive Attack Characteristics:** Passive attacks involve the attacker monitoring or observing system activity without actively altering it. The primary goal is to gather information or intelligence from the transmission. Examples include eavesdropping and traffic analysis, which are often very difficult to detect.
- **Active Attack Characteristics:** Active attacks involve the attacker actively disrupting or altering system or network activity. These attacks are focused on causing damage or modifying data in transit. Examples include masquerade, replay, modification of messages, and denial of service (DoS).
- **Security Mechanism Purpose:** A security mechanism is a process designed to detect, prevent, or recover from a security attack. These mechanisms are built into the system to identify any breaches of security. They are responsible for protecting systems against tampering and unauthorized access.
- **Encipherment Mechanism:** Encipherment, or encryption, uses algorithms to transform data into an unreadable form. This ensures that only individuals with the appropriate decryption key can read the information. It is the most common means of providing security for data in transit or storage.
- **Digital Signature Mechanism:** A digital signature is a cryptographic technique used to create a unique, verifiable identifier for a document. It ensures the authenticity and integrity of the digital message or file. This mechanism allows the receiver to verify that the sender is who they claim to be.
- **Security Service Definition:** A security service is a service that enhances the security of data processing and information transfers. These services are intended to counter security attacks by utilizing one or more mechanisms. They help in preventing potential risks to an organization's safety.
- **Authentication Service:** Authentication is the process of verifying the identity of a user or device before granting access. It ensures that the recipient can identify the sender and verify the source of the message. This service is fundamental to establishing trust in network communications.

- **Access Control Service:** Access control involves policies and procedures that determine who is allowed to access specific resources. It limits system access to authorized users and prevents unauthorized entities from reaching sensitive data. This service is crucial for maintaining internal security within an information system.

Q3. Provide a comprehensive overview of the Symmetric Cipher Model, its five essential ingredients, and two fundamental requirements.

Introduction: Symmetric encryption, also known as conventional or single-key encryption, uses the same key for both encryption and decryption. It is characterized by the use of a secret key shared between the sender and receiver.

- **Plaintext Ingredient:** Plaintext is the original intelligible message or data that is fed into the algorithm as input. It represents the information in its raw, readable form before any transformation occurs. All encryption processes begin with this fundamental component.
- **Encryption Algorithm Ingredient:** The encryption algorithm performs various substitutions and transformations on the plaintext. It uses mathematical logic to scramble the message into a form that is unreadable to opponents. The specific behavior of the algorithm is determined by the secret key.
- **Secret Key Ingredient:** The secret key is an input to the encryption algorithm that is independent of the plaintext and the algorithm. The algorithm produces different outputs depending on the specific key being used. It must be kept secret to ensure the security of the communication.
- **Ciphertext Ingredient:** Ciphertext is the scrambled, apparently random stream of data produced as output by the encryption algorithm. It depends entirely on the plaintext and the specific secret key used during the process. It is unintelligible to anyone who does not possess the correct key.
- **Decryption Algorithm Ingredient:** The decryption algorithm is essentially the encryption algorithm run in reverse. It takes the ciphertext and the same secret key to recover the original plaintext. This process ensures that the intended recipient can read the message.
- **Requirement for Strong Algorithm:** The first requirement is a strong encryption algorithm that can withstand cryptanalytic attacks. Even if an opponent knows the algorithm and has access to ciphertexts, they should be unable to figure out the key. The algorithm must be robust enough to prevent unauthorized decryption.
- **Requirement for Secure Key Exchange:** The second requirement is that the sender and receiver must obtain and keep the secret key in a secure fashion. If the key is discovered by an unauthorized party, all communications using that key become readable. Key distribution is one of the most significant challenges in symmetric systems.
- **Symmetric Dimensions:** Cryptographic systems are characterized by the type of operations used and the number of keys involved. Symmetric systems specifically use a single key for both parties. This distinguishes them from asymmetric systems that use public and private key pairs.
- **Substitution and Transposition:** Most symmetric algorithms are based on the principles of substitution and transposition. Substitution replaces elements in the plaintext with other elements, while transposition rearranges their order. Product systems often combine multiple stages of both for enhanced security.

- **Cryptanalyst Role:** A cryptanalyst attempts to recover the plaintext or the secret key by exploiting the characteristics of the algorithm. They may use sample plaintext-ciphertext pairs to deduce the key in use. The strength of the symmetric model is tested by its resistance to such attacks.

Q4. Detail the Playfair Cipher, including its matrix generation and the four specific rules for encryption.

Introduction: The Playfair Cipher was the first practical digraph substitution cipher, invented in 1854 by Charles Wheatstone. It encrypts pairs of letters (digraphs) rather than single characters, significantly improving security over monoalphabetic ciphers.

- **Key Square Generation:** The cipher uses a 5x5 grid of alphabets, known as a key square, which acts as the encryption key. The square is filled with the unique letters of a keyword, followed by the remaining letters of the alphabet in order. This matrix is the basis for all substitution rules in the cipher.
- **The I/J Rule:** Since a 5x5 matrix can only hold 25 letters, one letter (usually 'J') is omitted or combined with 'I'. If the plaintext contains 'J', it is replaced by 'I' before the encryption process begins. This ensures every letter in the plaintext can be mapped to the grid.
- **Digraph Splitting:** The plaintext is broken into pairs of two consecutive letters for processing. If the total number of letters is odd, a filler letter like 'Z' or 'X' is added to complete the final pair. This step is essential because the cipher only operates on letter pairs.
- **Repeated Letter Rule:** If a pair consists of two identical letters, a filler letter like 'X' is inserted between them. For example, the word "balloon" would be treated as "ba", "lx", "lo", "on". This prevents the encryption of identical letter pairs, which would weaken the cipher.
- **Rule 1 (Same Row):** If both letters in a digraph fall in the same row of the key matrix, each is replaced by the letter to its right. The rule is circular, so the first element of the row follows the last. For example, in a "MONARCHY" matrix, "AR" might encrypt to "RM".
- **Rule 2 (Same Column):** If both letters fall in the same column, each is replaced by the letter directly beneath it. Like the row rule, it is circular, so the top element follows the bottom. For example, "MU" might encrypt to "CM" based on their vertical positions.
- **Rule 3 (Rectangle Rule):** If the letters are in different rows and columns, they form the corners of a rectangle. Each letter is replaced by the letter in its own row but in the column occupied by the other letter of the pair. For example, "me" might be replaced by "gd".
- **Historical Significance:** The Playfair cipher was used for tactical purposes by British forces in World War I and by Australians in World War II. It was favored because it is fast to use and requires no special equipment. Its manual nature made it ideal for battlefield communications.
- **Security Strength:** Security is improved over monoalphabetic ciphers because there are 676 possible digraphs (26x26). Identification of individual letter frequencies is much more difficult for cryptanalysts. It requires a much larger frequency table for analysis.

- **Vulnerabilities:** Despite its strength, the Playfair cipher can be broken if a few hundred letters of ciphertext are known. Much of the plaintext structure is still retained in the ciphertext. Modern computational methods can easily crack it through statistical analysis.

Q5. Examine the Hill Cipher technique, focusing on its mathematical basis, encryption, and decryption processes.

Introduction: The Hill Cipher is a polygraphic substitution cipher based on linear algebra, proposed by Lester Hill. It processes blocks of plaintext letters as vectors and uses an invertible matrix as the encryption key.

- **Linear Algebra Basis:** The cipher is rooted in the principles of linear algebra and matrix multiplication. Each letter of the alphabet is represented by a number (typically A=0, B=1, ..., Z=25). Calculations are performed modulo 26 to keep the results within the alphabet range.
- **Block Processing:** Plaintext is broken into successive blocks of size 'm', where each block is treated as an n-component vector. The size of this vector must match the dimensions of the key matrix. This allows for the simultaneous encryption of multiple letters.
- **Key Matrix (K):** The key 'K' is a square m x m matrix of integers between 0 and 25. This matrix must be invertible modulo 26 to allow for successful decryption. The choice of the matrix determines the complexity and security of the cipher.
- **Encryption Formula:** To compute the ciphertext 'C', the plaintext vector 'P' is multiplied by the key matrix 'K'. The mathematical expression is $C = P \cdot k \text{ mod } 26$. This produces a new vector representing the encrypted characters.
- **Substitution Equations:** The substitution can be described as a system of 'm' linear equations. For a block size of 3, the equations involve multiplying plaintext values by matrix coefficients and summing them. This results in three distinct ciphertext values.
- **Decryption Formula:** To recover the original plaintext, the ciphertext vector 'C' is multiplied by the inverse of the key matrix k^{-1} . The formula is $P = c \cdot k^{-1}$. This operation reverses the transformation performed during encryption.
- **Finding the Inverse Matrix:** Decryption requires calculating the multiplicative inverse of the determinant of the key matrix modulo 26. The inverse matrix is then formed using the adjugate of the original matrix. Without the correct inverse, the plaintext cannot be recovered.
- **Strength Against Attacks:** The primary strength of the Hill cipher is that it completely hides single-letter frequencies. Because letters are encrypted in blocks, the statistical properties of individual letters are obscured. This makes it more secure than simple substitution ciphers.
- **Known-Plaintext Vulnerability:** Although strong against ciphertext-only attacks, the Hill cipher is easily broken with a known-plaintext attack. An attacker can set up a system of linear equations if they have enough plaintext-ciphertext pairs. This allows them to calculate the key matrix directly.
- **Vector Representation:** In practice, messages like "ACT" are converted into numerical vectors like before multiplication. The result is a new vector that corresponds to the ciphertext letters, such as "POH". This conversion process is standard for all Hill cipher operations.

5-Mark Questions

Q6. Compare and contrast Passive Attacks and Active Attacks with relevant examples.

Introduction: Security attacks are classified based on whether the attacker merely monitors data or actively attempts to modify system resources.

Feature	Passive Attack	Active Attack
Primary Goal	Gathering information or monitoring intelligence.	Causing damage, disruption, or altering data.
System Impact	No disruption or alteration of system activity.	Actively disrupts or alters system and network activity.
Detectability	Very difficult to detect as data remains unchanged.	Often noticeable due to deviations in message behavior.
Information Flow	Message is accessed but remains in its usual form.	Message is modified, delayed, or fabricated.
Examples	Eavesdropping and Traffic Analysis.	Masquerade, Replay, Modification, and DoS.

Q7. Identify and describe the four types of computer security as defined in the sources.

Introduction: Computer security is categorized into different types based on the specific domain or assets being protected.

1. **Cyber Security:** This involves securing computers, electronic devices, networks, and programs from attacks that occur when connected to the Internet. It specifically targets threats originating from global network connectivity.
2. **Information Security:** This type focuses on protecting a system's information from theft, illegal use, and piracy. Its primary objectives are to ensure the confidentiality, integrity, and availability of data.
3. **Application Security:** This means securing individual software applications and their associated data to prevent hacking. It ensures that application databases remain safe and private to the owner.
4. **Network Security:** This involves securing a network and protecting information about users connected through it. It guards against attacks like sniffing, spoofing, and man-in-the-middle attacks that target data packets.
5. **General Goal:** All these types aim to protect against unauthorized access, misuse, and information loss caused by hackers or invaders. They work together to ensure the overall safety of an organization's digital assets.

Q8. Explain the "Model for Network Security" and the four basic tasks involved in designing a security service.

Introduction: The Model for Network Security illustrates how information is transferred securely from one party to another over an insecure channel.

- **Security Transformation:** The model involves applying a security-related transformation, such as encryption, to the message to make it unreadable to opponents.
- **Secret Information:** A piece of secret information, such as an encryption key, is used in conjunction with the transformation to scramble and unscramble messages.
- **Trusted Third Party:** A third party may be required to distribute secret information or to arbitrate disputes regarding message authenticity.
- **Algorithm Design:** The first basic task is to design a robust algorithm for performing the security-related transformation that an opponent cannot defeat.
- **Key Management:** The remaining tasks include generating secret information, developing distribution methods, and specifying protocols for the principals to follow.

Q9. Discuss the Caesar Cipher, its mathematical algorithm, and its primary advantages.

Introduction: The Caesar Cipher is one of the earliest and simplest substitution ciphers, famously used by Julius Caesar for secret communications.

- **Shift Mechanism:** It works by shifting each letter in the plaintext by a fixed number of positions down the alphabet. For example, with a shift of 3, 'A' becomes 'D' and 'B' becomes 'E'.
- **Encryption Formula:** Mathematically, encryption is expressed as $C = E(k, p) = (p + k)$ where 'k' is the shift value between 1 and 25.
- **Decryption Formula:** Decryption is the reverse process, calculated as $p = D(k, C) = (C - k)$
- **Symmetric Nature:** It is a symmetric encryption technique, meaning the same fixed key is used by both the sender and the receiver.
- **Primary Advantages:** The cipher is very easy to implement, requires minimal pre-shared information, and serves as an excellent starting point for learning encryption.

Q10. Analyze the "Model for Network Access Security" and its two broad categories of defense.

Introduction: The Model for Network Access Security focuses on protecting computer systems and computing resources from unwanted access by human or software opponents.

- **Gatekeeper Function:** The first line of defense is a gatekeeper function designed to deny access to all but authorized users. This includes password-based login procedures and screening logic to reject viruses.
- **Internal Controls:** The second line of defense consists of internal controls that monitor system activity and analyze stored information. These controls attempt to detect the presence of intruders who have bypassed the gatekeeper.
- **Opponent Types:** The model protects against human opponents like hackers and software-based attacks like worms and viruses.
- **Access Channel:** It manages security across the access channel through which users and software interact with the information system.
- **System Resources:** The goal is to secure all computing resources, including the processor, memory, data, and software processes.

Q11. Describe the Vigenère Cipher and explain how it differs from a simple Caesar cipher.

Introduction: The Vigenère Cipher is a well-known example of a polyalphabetic substitution cipher that uses multiple substitution alphabets.

1. **Keyword Usage:** Unlike the Caesar cipher's single shift, the Vigenère cipher uses a repeating keyword to determine multiple shifts. Each letter of the keyword dictates a different Caesar shift for the corresponding plaintext letter.
2. **Encryption Process:** Encryption is performed using a Vigenère table or the formula $C_i = (P_i + K_i)$. The key must be as long as the message, usually achieved by repeating a shorter keyword.
3. **Multiple Ciphertexts:** A single plaintext letter can have multiple different ciphertext letters depending on its position relative to the keyword. For example, the same 'e' in "we are" might encrypt to different letters.
4. **Frequency Obscuration:** This method obscures individual letter frequency information, making it much harder to crack than a simple monoalphabetic shift.
5. **Strength vs Caesar:** While Caesar has only 25 possible keys, the Vigenère cipher's keyspace is determined by the length and variety of the keyword.

Q12. Explain the concept of a One-Time Pad and why it is considered unbreakable.

Introduction: The One-Time Pad is a scheme that uses a random key of the same length as the message to encrypt and decrypt a single communication.

- **Unbreakable Nature:** It is considered unbreakable because the ciphertext bears no statistical relationship to the original plaintext. It produces random output that contains no information about the message content.
- **Exhaustive Search Failure:** Even if an attacker performs an exhaustive search of all keys, they will find many legible plaintexts with no way to identify the correct one.
- **Key Randomness:** The absolute security of the scheme is derived entirely from the true randomness of the key used.
- **Single Use Policy:** To remain secure, each key must be used only once for a single message and then immediately discarded.
- **Practical Difficulties:** Despite its security, it faces challenges in generating massive quantities of random keys and the logistical problem of secure key distribution.

Q13. Describe the Rail Fence technique as a form of transposition cipher.

Introduction: The Rail Fence technique is the simplest form of a transposition cipher, which rearranges plaintext letters rather than substituting them.

- **Diagonal Writing:** To encrypt, the plaintext is written diagonally across a pre-determined number of rows (or "rails"). For example, the message is written in a zig-zag pattern.
- **Row Reading:** Once written, the ciphertext is generated by reading the letters off row by row. For a 2-row fence, all letters on the top row are followed by those on the bottom row.

- **Zero Substitution:** This technique does not change the actual letters used in the message; it only changes their positions.
- **Cryptanalysis Ease:** It is very easy to cryptanalyze because no complex key is involved and the original letter frequencies remain unchanged.
- **Enhanced Transposition:** Security can be improved by performing multiple stages of transposition, creating a more complex and harder-to-reconstruct permutation.

Q14. Explain the different types of cyber attacks, specifically focusing on Malware and Social Engineering.

Introduction: Cyber attacks are malicious actions that occur when a system is connected to the Internet, aiming to damage or steal data.

- **Malware Definition:** Malware (Malicious Software) is designed specifically to intrude into and damage computer systems. It includes various forms such as adware, spyware, ransomware, and Trojan horses.
- **Backdoor Entry:** A common malware tactic is the backdoor, where a malicious file is installed alongside legitimate free software to compromise security.
- **Social Engineering Definition:** Social engineering involves psychologically manipulating users to extract confidential data by gaining their trust.
- **Cognitive Exploitation:** Attackers exploit human cognitive biases and the natural trust people place in others to steal sensitive information.
- **Other Attacks:** These coexist with other threats like Denial of Service (DoS) attacks, which disrupt network availability by overwhelming resources with infinite requests.

Q15. Discuss the difference between Cryptanalysis and Brute-Force attacks.

Introduction: There are two general approaches to attacking a conventional encryption scheme: cryptanalysis and brute-force attacks.

- **Cryptanalysis Approach:** This relies on the nature of the encryption algorithm and knowledge of the general characteristics of the plaintext. It attempts to deduce the key or plaintext by exploiting mathematical weaknesses in the algorithm.
- **Brute-Force Approach:** The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is found. It does not require knowledge of the algorithm's flaws, only its existence.
- **Success Rate:** Success in cryptanalysis depends on the attacker's skill and available data, whereas brute-force success is guaranteed if given enough time and resources.
- **Average Effort:** For brute-force attacks, on average, half of all possible keys in the keyspace must be tried to achieve success.
- **Data Requirements:** Cryptanalysis may require sample plaintext-ciphertext pairs, while brute-force only requires the ciphertext and a way to recognize the valid plaintext.

2-Mark Questions

Q16. Define "Computer Security" according to the sources.

- **Core Practice:** Computer security is the practice of protecting computers, data, networks, and software from unauthorized access or misuse.
- **Preventative Focus:** It aims to shield hardware and information from theft, loss, and various security issues caused by hackers or invaders.

Q17. What is "Traffic Analysis" in the context of passive attacks?

- **Pattern Observation:** It involves analyzing network traffic patterns and metadata to gather intelligence about a system or network.
- **Non-Content Monitoring:** The intruder cannot read the encrypted message but understands the pattern, length, and frequency of communication.

Q18. Briefly explain the "Backdoor" attack.

- **Malicious Installation:** It occurs when malware or a virus is installed in a system, often concealed within otherwise useful free software.
- **Hidden Access:** The malicious file starts affecting the computer's security as soon as the main software is executed.

Q19. What is meant by "Data Integrity"?

- **Authorized Changes:** It is the assurance that information and programs are modified only in a specified and authorized manner.
- **Transmitted Safety:** It ensures that data remains untampered with while stored or during transmission in network packets.

Q20. Define "Non-repudiation" as a security service.

- **Denial Prevention:** It prevents a sender or receiver from denying a transmitted message occurred.
- **Verifiable Record:** It creates a verifiable record of the origin and transmission of data to resolve communication disputes.

Q21. What is "Masquerade" in active attacks?

- **Entity Falsification:** It is an attack where an intruder pretends to be an authentic sender to gain unauthorized access.
- **Credential Manipulation:** It often involves using stolen or forged credentials to bypass authentication controls.

Q22. Define "Traffic Padding" as a security mechanism.

- **Extra Data:** It is a technique of adding extra, meaningless data to a network traffic stream.
- **Obscuration Goal:** It aims to obscure the true content and volume of traffic, making it harder for attackers to perform traffic analysis.

Q23. What is a "Digital Signature"?

- **Unique Identifier:** It is a cryptographic mechanism that creates a unique, verifiable identifier for a digital document.
- **Trust Verification:** It is used to ensure the authenticity and integrity of a message, allowing the receiver to identify the sender.

Q24. Distinguish between Block Ciphers and Stream Ciphers.

- **Block Processing:** A block cipher processes input one fixed-size block of elements at a time, producing a corresponding output block.
- **Stream Processing:** A stream cipher processes input elements continuously, producing output one element at a time as it goes along.

Q25. What is "Spoofing"?

- **Masquerading Act:** Spoofing is the act of masquerading as a valid entity through the falsification of data like IP addresses or usernames.
- **Unauthorized Access:** It is performed to gain access to information or resources that the attacker is otherwise unauthorized to obtain.

NUMARICALS:

1. Caesar Cipher
2. Playfair Cipher
3. Hill Cipher
4. Vigenère Cipher
5. Rail Fence Technique
6. Vernam Cipher

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